

## FA2 / FA3 ATTENTION VS CUDNN SDPA

NVIDIA RTX PRO 5000 (SM-120A) B=1, H=32, N=16384, D=128

| CONFIGURATION       | TFLOPS | SPEEDUP | RELATIVE    |
|---------------------|--------|---------|-------------|
| CUDNN SDPA BASELINE | 232.4  | 1.00X   | <div></div> |

### ■ FA2 MMA

|              |       |       |             |
|--------------|-------|-------|-------------|
| SK=2, F16ACC | 157.9 | 0.68X | <div></div> |
| SK=1, F32ACC | 145.6 | 0.63X | <div></div> |
| SK=2, F32ACC | 166.5 | 0.72X | <div></div> |

### ■ FA2 CUTE MMA

|              |       |       |             |
|--------------|-------|-------|-------------|
| SK=1, F32ACC | 189.6 | 0.82X | <div></div> |
| SK=2, F32ACC | 197.0 | 0.85X | <div></div> |

### ■ FA2 TMA WS (1 WG)

|                    |       |       |             |
|--------------------|-------|-------|-------------|
| SK=1, SV=1, F16ACC | 161.2 | 0.69X | <div></div> |
| SK=2, SV=1, F16ACC | 189.4 | 0.81X | <div></div> |
| SK=2, SV=2, F16ACC | 190.5 | 0.82X | <div></div> |
| SK=2, SV=1, F32ACC | 204.9 | 0.88X | <div></div> |

### ■ FA3 TMA WS (2 WG)

|                    |       |       |             |
|--------------------|-------|-------|-------------|
| SK=1, SV=1, F16ACC | 210.1 | 0.90X | <div></div> |
| SK=1, SV=1, F32ACC | 210.8 | 0.91X | <div></div> |

### ■ FA2 CUTE TMA WS (1 WG)

|                    |       |       |             |
|--------------------|-------|-------|-------------|
| SK=2, SV=1, F32ACC | 220.0 | 0.95X | <div></div> |
| SK=3, SV=1, F32ACC | 223.4 | 0.96X | <div></div> |

### ■ FA2 CUTE TMA WS PERSISTENT-CTA (1 WG)

|       |       |       |             |
|-------|-------|-------|-------------|
| D=128 | 242.5 | 1.04X | <div></div> |
|-------|-------|-------|-------------|

## HGEMM CUTE SWIZZLE VS CUBLAS

NVIDIA RTX PRO 5000 (SM-120A) M=N=K=4096, F16ACC; FP8 4096/8192

| CONFIGURATION   | TFLOPS | SPEEDUP | RELATIVE    |
|-----------------|--------|---------|-------------|
| CUBLAS BASELINE | 236.9  | 1.00X   | <div></div> |

### ■ HGEMM CUTE SWIZZLE (F16ACC)

|               |       |       |             |
|---------------|-------|-------|-------------|
| S=2, BLK-SW=0 | 229.5 | 0.97X | <div></div> |
| S=2, BLK-SW=1 | 231.1 | 0.98X | <div></div> |
| S=3, BLK-SW=0 | 245.6 | 1.04X | <div></div> |
| S=3, BLK-SW=1 | 246.7 | 1.04X | <div></div> |

### ■ FP8 CUTE GEMM (F32ACC)

|                                  |       |       |             |
|----------------------------------|-------|-------|-------------|
| WS S2 128x256, 4096 <sup>3</sup> | 443.7 | 1.87X | <div></div> |
| WS S2 128x256, 8192 <sup>3</sup> | 487.8 | 2.06X | <div></div> |

SPEEDUP IS RELATIVE TO THE VENDOR BASELINE  
(CUBLAS FOR HGEMM, CUDNN SDPA FOR ATTENTION).

## FA SPLIT-D VS CUDNN SDPA

NVIDIA RTX PRO 5000 (SM-120A) B=1, H=32, N=8192, D=320

| CONFIGURATION       | TFLOPS | SPEEDUP | RELATIVE    |
|---------------------|--------|---------|-------------|
| CUDNN SDPA BASELINE | 69.8   | 1.00X   | <div></div> |

### ■ SPLIT-D TMA WS

|            |       |       |             |
|------------|-------|-------|-------------|
| SK=1, SV=1 | 86.6  | 1.24X | <div></div> |
| SK=2, SV=2 | 139.9 | 2.01X | <div></div> |

### ■ SPLIT-D TMA NON-WS

|            |       |       |             |
|------------|-------|-------|-------------|
| SK=2, SV=2 | 187.9 | 2.69X | <div></div> |
| SK=2, SV=3 | 188.8 | 2.71X | <div></div> |
| SK=3, SV=2 | 204.3 | 2.93X | <div></div> |

SK = K STAGES (FA), SV = V STAGES (FA), S = STAGES (HGEMM)